

Mean-Variance Portfolio Optimization

Active-Set Quadratic Programming with natID

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Numerical Optimization

Problem and Objective

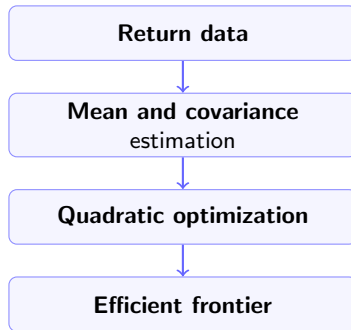
Goal

Construct a long-only portfolio that achieves a desired expected return while minimizing portfolio variance.

Given:

- estimated mean returns μ ,
- covariance matrix Σ ,
- target return r .

Application pipeline



Mathematical Formulation

For a target return r , the portfolio is obtained by solving

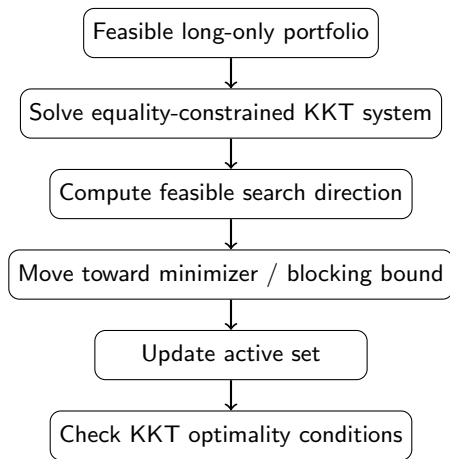
$$\begin{aligned} \min_w \quad & \frac{1}{2} w^T \Sigma w \\ \text{subject to} \quad & \mu^T w = r, \\ & \mathbf{1}^T w = 1, \\ & w \geq 0. \end{aligned}$$

- w – portfolio weights
- $w^T \Sigma w$ – portfolio variance
- $\mu^T w = r$ – target-return constraint
- $\mathbf{1}^T w = 1$ – full investment
- $w_i \geq 0$ – long-only constraint

The problem is a **convex quadratic program**.

Active-Set Quadratic Programming

The solver maintains a feasible portfolio and a working set of active zero-weight constraints.



If the optimality conditions are not satisfied, the process continues with the updated active set.

Implementation with natID

Numerical core

- CSV return-data parser
- mean-return estimation
- unbiased covariance estimation
- active-set QP solver
- efficient-frontier construction

natID integration

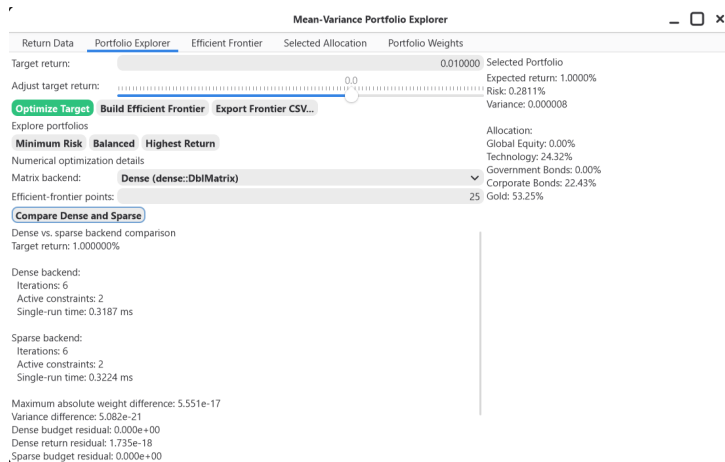
- `dense::DblMatrix`
- `sparse::DblSolver`
- identical KKT systems
- backend comparison
- natID graphical interface

The interface also exposes numerical information such as active-set iterations, active constraints and dense–sparse solution comparison.

Central workspace

- synchronized target-return controls
- selected portfolio summary
- frontier landmark navigation
- numerical backend details

Minimum Risk, Balanced, and Highest Return navigate already computed efficient-frontier solutions.

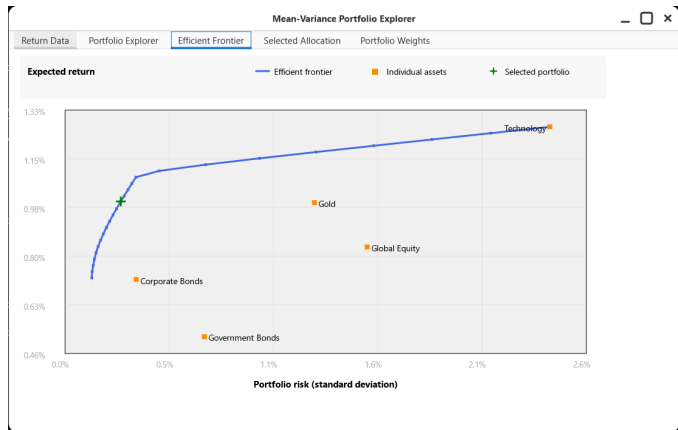


Efficient Frontier

Repeated target-return optimization produces the efficient frontier.

For every sampled target return, the application stores:

- optimal weights,
- achieved return,
- portfolio risk,
- variance,
- active constraints,
- solver iterations.

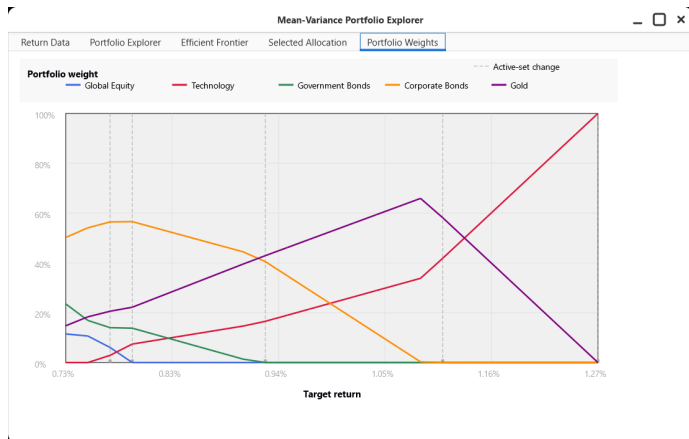


Portfolio Weights Across the Frontier

The weight-path visualization shows how the optimal allocation changes as the target return increases.

It also reveals:

- assets entering or leaving the portfolio,
- zero-weight constraints,
- changes in the active set,
- concentration at frontier extremes.



Validation and Demonstration Data

Numerical validation

- analytical two-asset case
- global minimum-variance case
- KKT-condition validation
- active non-negativity constraints
- infeasible-target rejection
- dense–sparse agreement
- 20-asset test systems

Data

- deterministic synthetic sample
- real-market demonstration sample
- 72 monthly observations
- January 2020–December 2025
- SPY, QQQ, IEF, LQD and GLD

The real-market data demonstrate the same optimization pipeline; they are not used as evidence of future investment performance.

- Implemented long-only mean-variance portfolio optimization as a convex quadratic program.
- Developed a feasible primal active-set solver with explicit KKT-based handling of active constraints.
- Constructed and visualized the efficient frontier and portfolio weight paths.
- Integrated professor-provided natID dense and sparse matrix backends.
- Verified the numerical implementation through automated tests and backend comparison.
- Demonstrated the complete pipeline on synthetic and historical market data.

Thank you